ReDeEP: Detecting Hallucination in Retrieval-Augmented Generation via Mechanistic Interpretability

A Reproduction and Robustness Study

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***Abstract*—Retrieval-augmented generation (RAG) is meant to ground a language model in retrieved evidence, yet models still hallucinate even when the evidence is correct. ReDeEP (Sun et al., ICLR 2025) detects such hallucinations by reading a model's internals: it measures how much *copying heads* use the retrieved context and how much *knowledge FFNs* inject parametric memory. We reproduce ReDeEP on the RAGTruth benchmark with a single 6 GB laptop GPU. On the paper's own backbone, Llama-2-7B-chat, our detector reaches an AUC of 0.739 against the published 0.7325, and the paper's two mechanistic findings replicate: all 1024 heads show lower context usage in hallucinated responses, and later-layer FFNs inject more parametric knowledge. Across six LLaMA-family models from 1.1B to 13B parameters, ReDeEP outperforms perplexity and entropy baselines on RAGTruth by 0.02 to 0.13 AUC, with Llama-2-13B the strongest detector (0.801), although per-task AUCs are markedly lower than the pooled figure. We then stress-test the method under corrupted retrieval. An irrelevant passage raises the hallucination score sharply, but corrupting the passage's numbers, so that the response contradicts it, leaves the score unchanged or lowers it: ReDeEP measures whether a response is *about* its context, not whether it *agrees* with it. Finally, prompted black-box judges from the Gemini and GPT families (AUC 0.827 and 0.790) outperform ReDeEP (0.721) on the same held-out responses, while a small 3B LLaMA judge performs at chance (0.499). Using the authors' released code and data, we also reproduce chunk-level ReDeEP (leakage-free AUC 0.731, identical to the authors' own logged scores), evaluate on Dolly (AC), and audit the official evaluation: selecting heads and layers on the test set inflates AUC by 0.016 on RAGTruth and 0.081 on Dolly (AC). With the released hyperparameters, the AARF mitigation's gate fires on every decoding step for the Llama-2 models and on only 8% for Llama-3-8B, and its apparent win over the baseline largely disappears once the judge's position bias is controlled.**

***Index Terms*—hallucination detection, retrieval-augmented generation, mechanistic interpretability, large language models, robustness, reproducibility**

I. Introduction

Large language models (LLMs) answer fluently but are prone to *hallucination*: generating content that is not supported by any evidence. Retrieval-augmented generation (RAG) is the standard remedy. A retriever fetches passages relevant to the user's query, and the model is instructed to answer from them. RAG is therefore widely deployed where trust matters, such as legal research, medical question answering and enterprise search.

RAG does not remove the problem. Even when the retrieved passage is accurate and contains the answer, models still produce claims that contradict or go beyond it. A model may be given a passage stating that a lunar mission returned 1,935.3 g of samples and still answer with a different number recalled from its training data. If a model can hallucinate when the correct answer is in front of it, adding retrieval is not a sufficient fix. We also need detectors that tell when a response has drifted away from its context.

ReDeEP [1] approaches this through *mechanistic interpretability*: rather than judging the output text, it inspects how the model produced it. Its central hypothesis is that two internal components compete. *Copying heads*, attention heads that move information from context tokens into the residual stream, carry the external knowledge. *Knowledge FFNs*, feed-forward layers in the later part of the network, inject the model's stored, parametric knowledge. A RAG hallucination occurs when the model under-uses the first and over-relies on the second. ReDeEP turns both effects into scores and combines them into a hallucination detector.

Our review of the method, prepared for this project, identified four gaps: it is validated only on LLaMA-family models, its head and layer selection is tuned by hand, its mitigation (AARF) gives only modest gains, and it *assumes the retrieved context is accurate*, never testing noisy or wrong retrieval, a common failure in deployed systems. This project reproduces the method and then targets the last gap directly.

Contributions.

1) A reproduction of ReDeEP's token-level detector on RAGTruth [2] on consumer hardware (RTX 4050, 6 GB VRAM). Memory optimisations make 7B models feasible, and the result matches the published AUC.
2) A replication of the paper's two mechanistic findings, and a transfer study across four LLaMA-family backbones [3] from 1.1B to 7B parameters, with ROC, per-task and ablation analyses.
3) A robustness extension that holds each response fixed and corrupts the retrieved passage, showing that ReDeEP

detects off-topic context but not contradicted facts.

4) A comparison against prompted black-box judges from the GPT, Gemini and LLaMA families on the same held-out responses.
5) Using the authors' released code and data: chunk-level ReDeEP, the Dolly (AC) dataset, the paper's two larger backbones (Llama-2-13B, Llama-3-8B), an audit of the official evaluation protocol, and an evaluation of the AARF mitigation with position-bias control.

II. Background and Related Work

A. RAG Hallucination versus Knowledge Conflict

ReDeEP distinguishes two failure modes. A *knowledge conflict* arises when the retrieved context itself is wrong or contradicts what the model knows. A *RAG hallucination* arises when the context is accurate and relevant, yet the response contradicts it or adds unsupported content. The paper targets only the second. Its datasets are chosen so that the context can be trusted, and the response is the only thing that can be wrong. Our robustness extension (Section IX) deliberately crosses this boundary to see how the method behaves just outside its stated scope.

B. Families of Hallucination Detectors

Sun et al. group prior detectors into three families by which signal they read and what confounds it:

- *Parametric confounded by external (PCE)* methods read hidden states to estimate what the model knows, for example EigenScore, SEP, SAPLMA and ITI. The external context mixes into these states and confounds the reading.
- *External confounded by parametric (ECP)* methods check the response against the context with prompts or retrieval-style checks, for example RAGAS, ChainPoll and RefChecker. The checking model's own knowledge leaks into the judgement.
- *Mixed parametric and external (MPE)* methods use uncertainty from sampling or token probabilities, for example perplexity, energy and LN-Entropy [4]. They blend both knowledge sources without separating them.

None of these *decouples* context usage from parametric-knowledge usage. ReDeEP's claim is that decoupling them gives a cleaner signal. In our experiments, perplexity and LN-Entropy represent the MPE family, and prompted LLM judges represent the ECP family.

C. Mechanistic Interpretability Tools

Transformer language models maintain a *residual stream*: each layer reads it, and its attention and FFN blocks add their outputs back into it. Two tools let ReDeEP read this stream. First, each attention head's *OV circuit* ($W_V W_O$) describes what information the head writes when it attends to a token. A head whose OV circuit maps a token's embedding back towards the same token's unembedding is a *copying head*. Second, the *logit lens* [7] applies the model's final norm and unembedding W_U to an intermediate residual state, giving the next-token distribution the model "would predict" at that depth. Prior work shows that FFN layers act as key-value memories that store factual knowledge [5], which motivates measuring what each FFN adds.

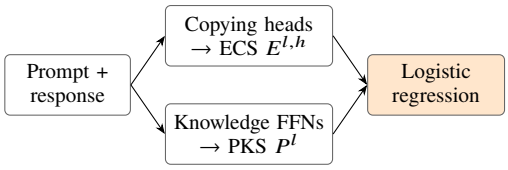


Fig. 1. ReDeEP pipeline: one forward pass yields an external-context score per head and a parametric-knowledge score per FFN layer; a logistic regression combines them into a hallucination score.

III. The ReDeEP Method

Fig. 1 summarises the detector. Given a prompt (instruction plus retrieved passage) and a response, ReDeEP runs one forward pass over their concatenation. Let R be the set of response token positions and C the set of context positions.

A. External Context Score (ECS)

For head h in layer l and response token n, let $\mathcal{T}_n \subset C$ be the 10% of context tokens that the head attends to most. With x_i^L the final-layer hidden state of token i,

$$E^{l,h} = \frac{1}{|R|} \sum_{n \in R} \cos\Big(\tfrac{1}{|\mathcal{T}_n|} \textstyle\sum_{i \in \mathcal{T}_n} x_i^L,\ x_n^L \Big). \tag{1}$$

ECS asks whether the content a head read is reflected in what the model wrote, not merely whether it attended to the right place. The paper finds that about 77.5% of hallucinations occur even though the heads attend to the correct context, so attention location alone is not enough.

B. Parametric Knowledge Score (PKS)

For layer l, the residual state just before the FFN block ($x^{\mathrm{mid},l}$) and just after the whole layer (x^l) are decoded with the logit lens. With $p_n = \sigma(W_U \,\mathrm{norm}(x_n))$ the resulting next-token distribution,

$$P^l = \frac{1}{|R|} \sum_{n \in R} \mathrm{JSD}\big(p_n^{\mathrm{mid},l},\ p_n^l\big), \tag{2}$$

where σ is the softmax and JSD the Jensen–Shannon divergence. A large divergence means the FFN changed what the model wanted to say, so it injected its own knowledge.

C. Selecting Copying Heads and Knowledge FFNs

Copying heads are found from weights alone. For each head we form the matrix $M = W_E\, W_V^h W_O^h\, W_U^\top$ restricted to a fixed random sample of 500 vocabulary tokens. A copying head has a large-magnitude trace, meaning tokens map strongly onto themselves, and few Gershgorin-disc outliers, meaning a well-conditioned diagonal. Heads are ranked by the sum of both ranks, and the top $|\mathcal{A}| = 8$ form the set $\mathcal{A}$. Knowledge FFNs are chosen with labels. On each training fold, the $|\mathcal{F}| = 5$ layers whose PKS correlates most positively with the hallucination label form the set $\mathcal{F}$.

Algorithm 1 ReDeEP scoring of one response

Require: prompt c, response r, heads $\mathcal{A}$, layers $\mathcal{F}$, fitted regression g

run one forward pass on $[c; r]$; hooks record per layer the top-k attended context indices and $x^{\text{mid},l}$
for each head $(l, h) \in \mathcal{A}$ **do**
    $E^{l,h} \leftarrow$ mean over $n \in R$ of $\cos(\text{mean}_{i \in \mathcal{T}_n} x_i^L,\ x_n^L)$
end for
for each layer $l \in \mathcal{F}$ **do**
    $P^l \leftarrow$ mean over $n \in R$ of $\text{JSD}(p_n^{\text{mid},l}, p_n^l)$
end for
return $g\big(\text{mean}_{\mathcal{F}} P,\ -\text{mean}_{\mathcal{A}} E\big)$

D. Detector

ReDeEP models the hallucination score as $H = \alpha\, \bar{P}_{\mathcal{F}} - \beta\, \bar{E}_{\mathcal{A}}$, with $\alpha, \beta > 0$, where the bars denote means over the selected layers and heads. We fit this as a logistic regression on the two standardised features $\bar{P}_{\mathcal{F}}$ and $-\bar{E}_{\mathcal{A}}$. Its probability output is the score, and the decision threshold is the best-F1 point on the training fold. Algorithm 1 gives the scoring procedure.

E. AARF Mitigation

The paper also proposes AARF ("Add Attention, Reduce FFN"). At decoding time it scales copying-head outputs by $\alpha_2 > 1$ and knowledge-FFN outputs by $\beta_2 < 1$, steering generation towards the context without fine-tuning. In the released code the intervention is gated per token: it is applied only when the token-level ReDeEP score, min-max normalised with stored ranges, exceeds a threshold (0.6, with $\alpha_2 = 1.2$, $\beta_2 = 0.8$). We port this gate faithfully and evaluate AARF in Section X.

F. Chunk-Level ReDeEP

ReDeEP(chunk) works on text chunks instead of single tokens. Prompt and response are split into chunks (the released `prompt_spans` and `response_spans`). For each response chunk and copying head, ECS is the BGE [6] embedding similarity between the response chunk and the prompt chunk that receives the most attention mass from it under that head. PKS is the token-summed JSD of each layer over the chunk. The authors use their own top-32 copying heads per model, which they release; we use the same heads. Our hooks keep only these heads' chunk-level attention sums, so long prompts fit in 6 GB.

IV. Dataset

A. RAGTruth

RAGTruth [2] is the first large word-level hallucination corpus built specifically for RAG. It contains 17,790 responses to 2,965 source prompts. Each response was generated by one of six LLMs: GPT-3.5-turbo, GPT-4, Llama-2-7B/13B/70B-chat and Mistral-7B-Instruct. Human annotators marked every hallucinated span with its character offsets and type. It covers three task types: question answering over MS MARCO passages, summarisation of news articles, and data-to-text generation from structured Yelp business records.

TABLE I
RAGTruth by task: full corpus and our evaluation subset

		Full corpus		Our subset	
Task	Source	Resp.	Halluc.	Resp.	Halluc.
Question answering	MS MARCO	5,934	29%	149	35%
Summarization	CNN/DM, news	5,658	30%	121	35%
Data-to-text	Yelp	6,198	69%	145	81%

TABLE II
Backbones

Backbone	Params	Layers×Heads	Vocab	Precision
TinyLlama-1.1B-Chat	1.1B	22×32	32k	fp16
Llama-3.2-1B-Instruct	1.2B	16×32	128k	fp16
Llama-3.2-3B-Instruct	3.2B	28×24	128k	4-bit NF4
Llama-2-7B-chat	6.7B	32×32	32k	4-bit NF4

The corpus has two files. `source_info.jsonl` holds each source's identifier, task type and full prompt including the retrieved context. `response.jsonl` holds each response with its generator model, train/test split, a quality flag for refusals or truncation, and the list of hallucination spans. Following the paper, a response is labelled hallucinated if it contains at least one span. We keep responses flagged as good quality.

Evaluation subset. The paper evaluates each backbone on responses that backbone generated. We use the Llama-2-7B-chat setting: all good-quality *test-split* responses generated by llama-2-7b-chat, capped at 6,000 characters of prompt plus response so that every example fits in 1,024 tokens. This yields 415 responses, of which 212 (51%) are hallucinated. Table I breaks them down by task.

B. Dolly (AC)

The paper also evaluates on Dolly (AC), a subset of Databricks' dolly-15k instruction data filtered to prompts with an accurate context passage, answered by each backbone and hand-labelled. The authors' released dataset contains it: 100 question-answering prompts with responses from Llama-2-7B, Llama-2-13B and Llama-3-8B (99 test responses each). Dolly (AC) carries only a response-level label (hallucinated or not), with no span positions, so we evaluate it at the response level. The release also includes Llama-3-8B-Instruct's own RAGTruth responses (450 test responses) and the chunk boundaries used by chunk-level ReDeEP.

V. Experimental Setup

A. Backbones

Table II lists the four LLaMA-architecture backbones. All share the pre-norm decoder with RMSNorm, rotary embeddings and SwiGLU FFNs, so the same code applies to each. Llama-3.2 uses grouped-query attention, which our copying-head analysis handles by pairing each query head with its shared key-value head. Weights are loaded from public re-uploads of the official releases, so no gated licence is required.

B. Hardware and Software

All experiments ran on a Lenovo LOQ laptop with an NVIDIA RTX 4050 Laptop GPU (6 GB VRAM) and 16 GB RAM, under Windows 11. We used PyTorch 2.11 with CUDA 12.8, Hugging Face transformers 5.17 and bitsandbytes 0.50. The 3B and 7B models were loaded in 4-bit NF4 with fp16 computation; the smaller models ran in fp16.

C. Engineering for a 6 GB GPU

The reference pipeline asks the model to return every full attention matrix. For Llama-2-7B at 1,024 tokens that is 32 layers $\times$ 32 heads $\times$ 1024^2 values, which does not fit beside the weights. We made three changes:

1) Forward hooks on each attention module keep only the top-k context indices per head and response token, then discard the attention map.
2) The mean of attended hidden states is computed as a sparse averaging-matrix product instead of gathering a $[H, |R|, k, d]$ tensor.
3) Perplexity and LN-Entropy reuse the logits of the same forward pass instead of a second pass.

On a RAGTruth example, the rewritten ECS matched the original implementation to within 6×10^{-8}, so results remain comparable. Copying-head weights are dequantised from 4-bit on the fly. Prompts longer than the budget are truncated from the left, so the whole response is always scored.

D. Evaluation Protocol

We report AUC (threshold-free ranking quality), the Pearson correlation (PCC) between score and label, and recall and F1 at the training-fold threshold. Our main protocol is stratified 5-fold cross-validation over the 415 responses, reported as mean ± standard deviation over folds. Copying heads are fixed from weights, and knowledge FFNs and the regression are re-fitted inside each training fold, so no test label influences selection. For comparisons with the judges we also use a single fixed 70/30 split, whose test part has 124 responses.

E. Baselines

Perplexity is the exponentiated mean negative log-likelihood of the response tokens given the prompt. **LN-Entropy** is the mean per-token entropy of the predictive distribution over the response [4]. Both come from the same forward pass and backbone as ReDeEP. Higher values are taken to indicate hallucination.

F. Black-Box Judges

Each judge receives the RAG input and the response and replies in JSON with the probability (0–100) that the response contains a hallucination (prompt in Appendix B). We used Gemini 3.5 Flash-Lite through the Google AI Studio free tier. For the GPT family we used OpenAI's open-weights gpt-oss-120b, served free by Groq. We also ran Llama-3.2-3B-Instruct locally in 4-bit. The stated probability is the score.

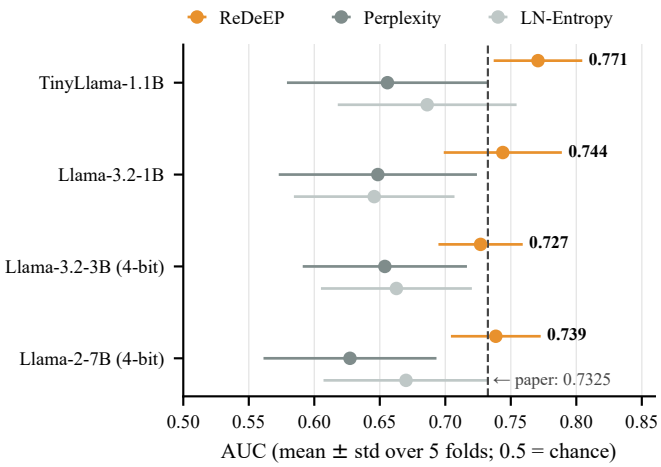


Fig. 2. Detection AUC per backbone (mean ± std over 5 folds). ReDeEP (orange) beats both baselines on every model; the dashed line marks the paper's Llama-2-7B result.

TABLE III
DETECTION RESULTS, 5-FOLD CV MEAN (n = 415)

	ReDeEP			PPL	LN-Ent
Backbone	AUC	PCC	F1	AUC	AUC
TinyLlama-1.1B	**0.771**	0.475	0.718	0.656	0.686
Llama-3.2-1B	**0.744**	0.426	0.692	0.649	0.646
Llama-3.2-3B (4-bit)	**0.727**	0.399	0.695	0.654	0.663
Llama-2-7B (4-bit)	**0.739**	0.427	0.687	0.627	0.670
Paper (Llama-2-7B)	0.7325	0.3979	–	–	–

VI. RESULTS

A. Hallucination Detection

ReDeEP reproduces on the paper's backbone: Llama-2-7B reaches an AUC of 0.739 (±0.034) against the published 0.7325, and a PCC of 0.427 against 0.3979 (Table III, Fig. 2). It beats both baselines on every backbone. The margin over the better baseline ranges from 0.06 (Llama-3.2-3B against LN-Entropy) to 0.10 (Llama-3.2-1B against perplexity). Detection quality is roughly flat in model size: the 1.1B TinyLlama scores highest, and the 3B and 7B models lie within one standard deviation of each other. ReDeEP is also more stable across folds (std 0.03–0.05) than the baselines (0.06–0.08).

Fig. 3 shows ROC curves built from out-of-fold predictions. ReDeEP dominates the baselines over most of the false-positive range rather than only at one operating point. Fig. 4 shows that its scores separate the two classes but overlap heavily in the middle, which matches an AUC in the mid-0.7s.

B. Replicating the Mechanistic Findings

The paper reports two core observations on Llama-2-7B: copying heads use less context in hallucinated responses (1006 of 1024 heads), and later-layer FFNs inject more parametric knowledge. Both replicate (Table IV). On Llama-2-7B, all 1024 of 1024 heads have higher mean ECS on truthful responses (Fig. 5). PKS is higher on hallucinated responses

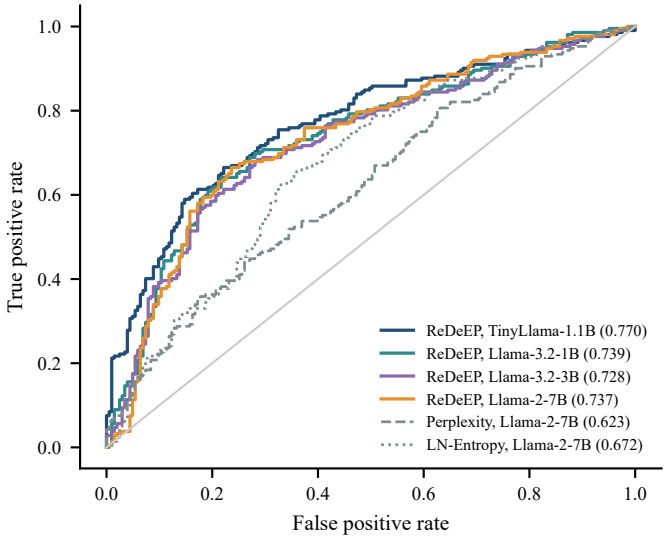


Fig. 3. ROC curves from out-of-fold scores. All four ReDeEP curves lie above the perplexity and LN-Entropy curves of Llama-2-7B across most of the range.

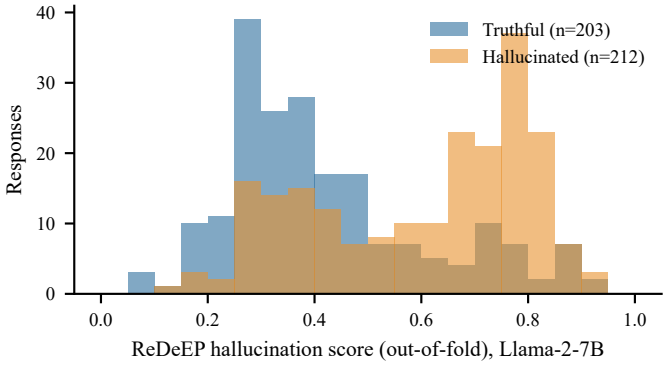


Fig. 4. Distribution of out-of-fold ReDeEP scores on Llama-2-7B. Hallucinated responses shift right, with substantial overlap in the middle.

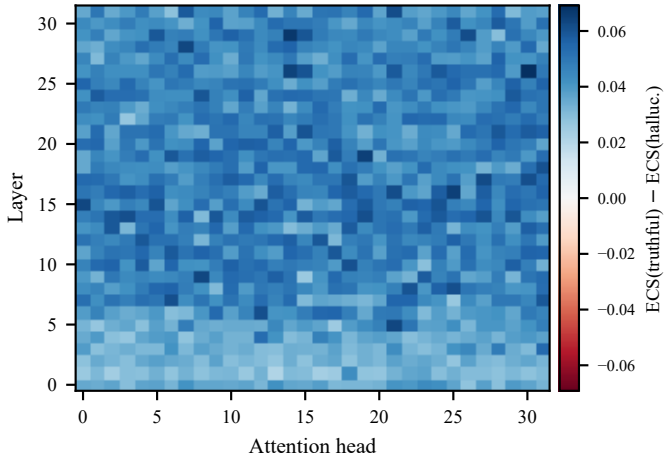


Fig. 5. Mean ECS of truthful minus hallucinated responses for every head of Llama-2-7B. Every cell is positive: context usage is lower in hallucinated responses across all heads.

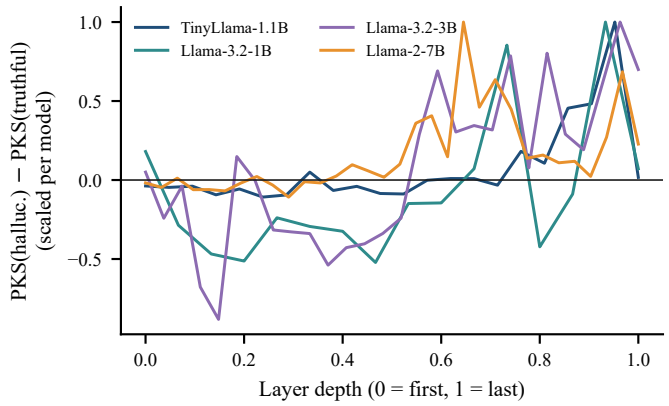


Fig. 6. Mean PKS of hallucinated minus truthful responses by relative layer depth. The difference is concentrated in the later layers, where knowledge FFNs sit.

in 16 of the 16 later layers, but only 6 of the 16 earlier ones (Fig. 6). TinyLlama and Llama-3.2-3B show the same pattern clearly; Llama-3.2-1B shows it more weakly (4 of 8 later layers against 1 of 8 earlier ones).

One caution applies to the ECS result. Because ECS compares final-layer hidden states for every head, the per-head scores of a response are strongly correlated. The near-unanimous head count therefore reflects a shift at the level of the whole response more than 1024 independent effects.

C. Performance by Task

Pooled AUC hides large differences between tasks (Table V, Fig. 7). Within a single task, ReDeEP's AUC falls to between 0.40 and 0.66. On Llama-2-7B it is 0.623 on QA, 0.561 on summarisation and 0.396 on data-to-text. ReDeEP beats the baselines on QA for every backbone, but the baselines win on summarisation for three of the four models, and on data-to-text ReDeEP is near or below chance for the three Llama models.

Part of the pooled score therefore comes from separating *tasks*, not responses. The hallucination rate differs strongly by task in our subset: 35% for QA, 35% for summarisation and 81% for data-to-text. Internal signals also differ systematically by task. Data-to-text prompts are structured JSON, whose hidden states plausibly look unlike prose, so a detector may partly identify the task and inherit its base rate. The paper reports pooled numbers too, so the comparison with it is fair, but within-task evaluation is the more demanding test for deployment.

D. Ablations

Both signals matter. Table VI fits the detector on each feature alone. On Llama-2-7B, ECS alone reaches 0.711 and PKS alone 0.708, while combining them gives 0.737. The combination is the best on every backbone, which supports ReDeEP's central claim that context usage and parametric injection are complementary. Which signal is stronger depends

TABLE IV
REPLICATION OF THE PAPER'S TWO MECHANISTIC FINDINGS

Backbone	$L \times H$	Heads: ECS truth > halluc.	Layers: PKS halluc. > truth Early	Late
TinyLlama-1.1B	22×32	704/704	1/11	8/11
Llama-3.2-1B	16×32	512/512	1/8	4/8
Llama-3.2-3B	28×24	672/672	2/14	13/14
Llama-2-7B	32×32	1024/1024	6/16	16/16
Paper (Llama-2-7B)	32×32	1006/1024	higher in later layers	

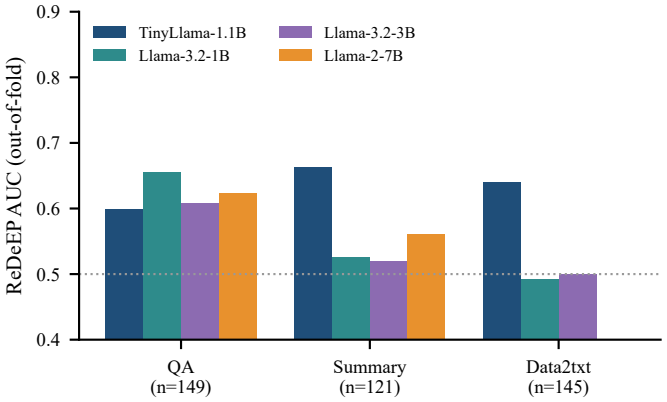


Fig. 7. Out-of-fold ReDeEP AUC within each task type. Dotted line: chance.

on the model: PKS dominates on TinyLlama, and ECS on the Llama-3.2 models.

Sensitivity to $|\mathcal{A}|$ and $|\mathcal{F}|$. One of the gaps we identified was that head and layer selection is tuned by hand. Fig. 8 sweeps 1 to 32 copying heads and 1 to 10 FFN layers on Llama-2-7B. AUC stays between 0.736 and 0.751 across the whole grid, against 0.737 for our default. The detector is therefore insensitive to these hyperparameters in our setting, and the hand-tuning concern matters less than expected. This is consistent with the correlated per-head ECS noted above.

E. Computational Cost

Table VII reports wall-clock costs. Copying heads are computed once per model from weights in a few seconds. Scoring takes a median of about half a second per response for TinyLlama and 4.5 s for Llama-2-7B in 4-bit. The complete Llama-2-7B run over 415 responses took 38 minutes on a laptop. ReDeEP needs no sampling, no second model and no network access.

VII. Reproduction with the Official Code and Data

The authors released their code, the selected copying heads and hyperparameters for each model, the Dolly (AC) and Llama-3-8B data, and the chunk-level scores they computed for Llama-2-7B. This let us reproduce the remaining parts of the paper and check our numbers directly against theirs.

TABLE V
AUC WITHIN EACH TASK: REDEEP / BEST BASELINE (BOLD: REDEEP WINS)

Backbone	QA	Summary	Data-to-text
TinyLlama-1.1B	**0.598** / 0.582	**0.662** / 0.659	**0.639** / 0.458
Llama-3.2-1B	**0.655** / 0.563	0.525 / 0.668	**0.492** / 0.419
Llama-3.2-3B	**0.608** / 0.552	0.519 / 0.680	**0.499** / 0.465
Llama-2-7B	**0.623** / 0.539	0.561 / 0.652	0.396 / 0.505

TABLE VI
FEATURE ABLATION: OUT-OF-FOLD AUC

Backbone	ECS only	PKS only	Both
TinyLlama-1.1B	0.727	0.749	**0.770**
Llama-3.2-1B	0.717	0.660	**0.739**
Llama-3.2-3B	0.717	0.635	**0.728**
Llama-2-7B	0.711	0.708	**0.737**

A. Token- and Chunk-Level Results

Table VIII collects the results. Token-level RAGTruth numbers use our protocol from Section VI on each backbone's own responses; chunk-level numbers use the official chunks and copying heads. For chunk-level ReDeEP we report both the leakage-free nested cross-validation and, in brackets, the official protocol described in Section VII-B.

Chunk level matches the authors exactly. Our chunk-level features agree with the authors' logged features head by head in 148 of 160 cases on the responses we spot-checked, and our PKS correlates at 0.994 with theirs. With leakage-free evaluation, our Llama-2-7B chunk-level AUC is 0.731, against 0.731 from the authors' own logged scores under the same protocol. Under the official protocol we obtain 0.740 against their 0.747; the small gap is consistent with our 4-bit weights.

Dolly (AC). On the second dataset ReDeEP(token) reaches 0.690 on Llama-2-7B, against 0.452 for perplexity and 0.615 for LN-Entropy. With only 99 responses, fold-to-fold variation is large (±0.096), so these numbers are less precise than RAGTruth's.

Larger backbones. Llama-2-13B-chat and Llama-3-8B-Instruct do not fit a 6 GB GPU, so we ran them in 4-bit on a free Kaggle T4 GPU with the same code, each scoring its own responses. Llama-2-13B reaches a token-level AUC of 0.801 on its own 450 RAGTruth responses (perplexity 0.553, LN-Entropy 0.667), 0.794 at chunk level, and 0.646 on Dolly (AC) (LN-Entropy 0.555); Llama-3-8B reaches a token-level AUC of 0.730 on its own 450 RAGTruth responses (perplexity 0.661, LN-Entropy 0.712), 0.717 at chunk level, and 0.538 on Dolly (AC) (LN-Entropy 0.641). Llama-2-13B is the strongest detector we tested. On Llama-3-8B, ReDeEP's margin over LN-Entropy is small on RAGTruth and reversed on Dolly (AC), consistent with the paper's observation that heads and layers must be re-tuned per model; Llama-3-8B's released labels are response-level only, so its chunk-level score is evaluated per response.

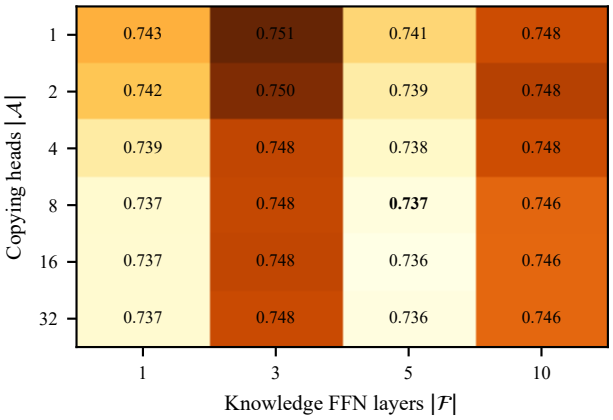


Fig. 8. Out-of-fold AUC on Llama-2-7B as the number of copying heads and knowledge-FFN layers varies. Bold: the default (8 heads, 5 layers).

TABLE VII
COMPUTE COST ON AN RTX 4050 (415 RESPONSES)

Backbone	Copying heads (s)	Per response (s)	Total (min)
TinyLlama-1.1B	3.7	0.5	3.7
Llama-3.2-1B	2.6	0.7	5.0
Llama-3.2-3B	4.8	1.4	9.6
Llama-2-7B	7.5	4.5	37.9

B. Auditing the Official Evaluation Protocol

The released regression scripts rank the 32 copying heads and the FFN layers by their AUC *on the evaluation set itself*, keep the best ones, and combine them with weights tuned per model and dataset; the resulting AUC is reported on the same responses. Test labels therefore influence both feature selection and hyperparameters. To measure the effect, we re-evaluated the authors' own logged chunk scores and our own runs in three ways (Table IX).

We first confirmed our re-implementation of the official regression: on the authors' logged scores it returns exactly their stored result (AUC 0.747, PCC 0.421). Tuning the hyperparameters on the test set as well (the "all tuned" column, an upper bound of the leak) raises the AUC further, while moving feature selection and hyperparameter choice into the training folds (nested CV) lowers it. On RAGTruth (450 responses) the inflation is small, 0.016 AUC, so the paper's main conclusions stand. On Dolly (AC) (99 responses) it reaches 0.081 for Llama-2-7B, and a similar inflation of about 0.08 appears for Llama-2-13B and Llama-3-8B: with few responses, choosing the best of 32 heads and 32 layers on the test set fits noise. Reported results on small benchmarks should therefore use held-out selection.

VIII. Comparison with Black-Box Judges

Large judges win on accuracy. Gemini 3.5 Flash-Lite reaches 0.827 AUC and gpt-oss-120b 0.790, against 0.721 for ReDeEP on the same 124 responses (Table X, Fig. 9). On all 415 responses Gemini scores 0.810, consistent with the held-out result. The judges read the passage and the response as text, so they can directly spot a number or name that does not match, which is exactly the weakness Section IX exposes in ReDeEP. Both judges have very high recall (0.91–0.93): they flag most hallucinations, at the cost of more false alarms.

TABLE VIII
REDEEP AUC WITH THE OFFICIAL DATA, LEAKAGE-FREE NESTED 5-FOLD CV (OFFICIAL-PROTOCOL AUC IN BRACKETS)

	RAGTruth		Dolly (AC)	
Backbone	Token	Chunk	Token	Chunk
Llama-2-7B	0.739	0.731 (0.740)	0.690	0.713 (0.795)
Llama-2-13B	0.801	0.794 (0.795)	0.646	0.683 (0.768)
Llama-3-8B	0.730	0.717 (0.727)	0.538	0.634 (0.717)

TABLE IX
EFFECT OF SELECTING FEATURES ON THE TEST SET (CHUNK LEVEL)

Scores	Official protocol	All tuned on test	Nested CV (ours)
Authors' logs, RAGTruth	0.747	0.747	**0.731**
Ours, RAGTruth	0.740	0.745	**0.731**
Ours, Dolly (AC)	0.795	0.805	**0.713**
Ours Llama-2-13B, RAGTruth	0.795	0.802	**0.794**
Ours Llama-2-13B, Dolly (AC)	0.768	0.832	**0.683**
Ours Llama-3-8B, RAGTruth	0.727	0.732	**0.717**
Ours Llama-3-8B, Dolly (AC)	0.717	0.749	**0.634**

Judge scores are coarse. Although asked for a probability, Gemini answered exactly 0 or 100 for 99.5% of responses, which turns its output into a near-binary verdict. gpt-oss-120b used 17 distinct values. A finer-grained score would likely raise both judges' AUC further.

Small judges fail. Llama-3.2-3B answered 0 or 100 for 99.8% of responses and almost always said "no hallucination", catching 7% of hallucinations (AUC 0.499). Reading the internals of a model of the same size (ReDeEP on Llama-3.2-3B, AUC 0.727) works far better than asking it.

Trade-off. The judges rely on large models served behind an API, with per-call cost, rate limits and data leaving the machine. Our free-tier access was limited to 20 requests per day for the newest Gemini Flash model, which forced the use of Flash-Lite. ReDeEP runs locally on a laptop with a 7B open model. The practical choice depends on whether accuracy or cost and privacy dominate.

IX. Extension: Robustness Under Corrupted Retrieval

ReDeEP assumes accurate context. We ask what its signals measure when that assumption is broken on purpose. Two outcomes are possible. If ECS and PKS track *faithfulness to the given context*, a response should look hallucinated whenever its context stops supporting it. If they track something weaker, such as topical similarity, some kinds of contradiction will go unnoticed.

A. Design

We hold each response fixed and change only the retrieved passage, under three conditions:

1) *Accurate:* the real RAGTruth passage (control).

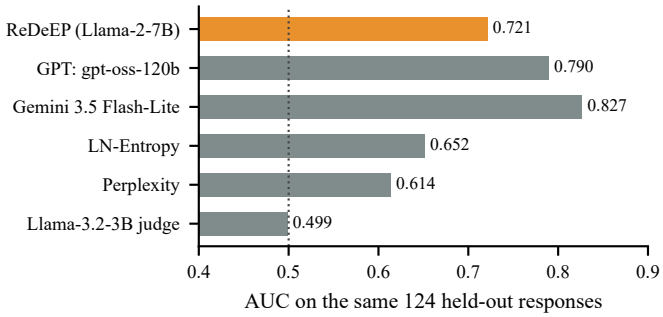


Fig. 9. All detectors on the same 124 held-out responses. The API judges are more accurate; a small local judge is at chance (dotted line, AUC 0.5).

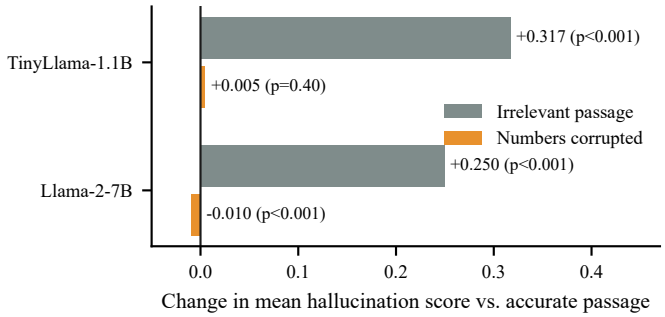


Fig. 10. Change in mean ReDeEP score relative to the accurate passage (paired Wilcoxon test, 150 responses per backbone).

TABLE X
JUDGES VS. REDEEP ON THE SAME 124 HELD-OUT RESPONSES

Detector	Kind	AUC	PCC	Rec.	F1
Gemini 3.5 Flash-Lite	Judge	**0.827**	0.671	0.914	0.865
GPT: gpt-oss-120b	Judge	0.790	0.465	0.929	0.778
ReDeEP (Llama-2-7B)	Internals	0.721	0.390	0.714	0.714
LN-Entropy	Uncertainty	0.652	–	–	–
Perplexity	Uncertainty	0.614	–	–	–
Llama-3.2-3B (local)	Judge	0.499	-0.005	0.071	0.127

TABLE XI
ROBUSTNESS: MEAN SCORE AND SIGNALS PER CONDITION

Backbone	Condition	Score	ECS	PKS	AUC
TinyLlama	Accurate	0.522	0.326	0.158	0.755
TinyLlama	Irrelevant	0.840	0.244	0.187	0.545
TinyLlama	Numbers corr.	0.528	0.330	0.160	0.747
Llama-2-7B	Accurate	0.521	0.282	0.091	0.710
Llama-2-7B	Irrelevant	0.770	0.288	0.114	0.564
Llama-2-7B	Numbers corr.	0.510	0.310	0.092	0.715

2) *Irrelevant:* a passage from a different source of the same task type. Nothing in the response is supported any more.
3) *Numbers corrupted:* every number in the real passage is multiplied by a random factor in $[1.3, 3.0]$, with a fixed seed per example. Numbers the response copied are now contradicted by the passage, while the topic and wording are unchanged.

We draw 150 responses (75 hallucinated), refit the detector on the remaining 265, and score each response under all three passages with TinyLlama-1.1B and Llama-2-7B. A faithful detector should raise its score under both corrupted conditions. We test the per-response score change with a paired Wilcoxon signed-rank test.

B. Results

Irrelevant passages work as intended. The score jumps for nearly every response, by +0.317 for TinyLlama and +0.250 for Llama-2-7B (both $p < 0.001$). The original labels then stop separating the responses (AUC ≈0.55; Fig. 10, Table XI). This is correct, because nothing in a response is supported by the new passage. The mechanism differs by size. In TinyLlama, ECS falls from 0.326 to 0.244, as copying heads find nothing useful to copy. In Llama-2-7B, ECS is flat and the increase comes from PKS (0.091 to 0.114): the larger model falls back on its own knowledge.

Corrupted numbers go unnoticed. For TinyLlama the score does not move (+0.005, $p = 0.40$). For Llama-2-7B it *falls* slightly (-0.010, $p < 0.001$) while ECS rises from 0.282 to 0.310, so the detector rates a contradicted response as more faithful. The corruption may not touch every response, so we checked the 122 responses that state a number the corrupted passage now contradicts. Their score rose in only 52% of cases for TinyLlama, which is chance, and 31% for Llama-2-7B. Detection AUC against the original labels is also unchanged under this condition, so the detector behaves as if the passage were still accurate.

C. Interpretation

ECS is a cosine similarity between hidden states. It measures whether the response is built from the same *content* the copying heads read, not whether individual facts agree. A changed number in an otherwise identical sentence plausibly moves the hidden state very little, so ReDeEP cannot see it. PKS does not compensate: the FFNs inject no more parametric knowledge when the passage's numbers change.

ReDeEP is therefore a detector of *topical* hallucination: a response that drifts away from what the context is about. It is not a detector of *factual* inconsistency with the context. In deployment, it should be paired with a fact-level check, such as matching numbers and entities between response and context, or a text-level judge. Notably, the paper's motivating example is itself a wrong number, which is the kind of error our experiment suggests ReDeEP is least sensitive to.

X. AARF MITIGATION

A. Setup

Following the released scripts, we generated greedy answers to the 99 Dolly (AC) test prompts with Llama-2-7B, once normally and once with gated AARF, using the authors' Llama-2-7B Dolly hyperparameters (4 heads, 3 layers, weight 0.2, $\alpha_2 = 1.2$, $\beta_2 = 0.8$, threshold 0.6). The authors judged each differing pair with GPT-4o, always showing the AARF answer first and asking which is more truthful. We use the

TABLE XII
AARF VS. BASELINE ON DOLLY (AC): ANSWERS CHANGED, GATE FIRING RATE, AND JUDGE WINS (AARF / BASELINE)

Backbone	Changed answers	Gate fires	Official order	Both orders agree
Llama-2-7B	63/99	100%	33 / 24	19 / 15
Llama-2-13B	54/98	100%	30 / 20	14 / 12
Llama-3-8B	2/99	8%	2 / 0	0 / 0

same prompt with Gemini 3.5 Flash-Lite (free tier). We judge every pair in *both* orders, because LLM judges are known to favour one position.

B. The Gate Always Fires

We logged the gate score at every decoding step. For Llama-2-7B, across all 10,161 steps, the normalised score never fell below 0.83 (median 1.26), so the 0.6 threshold was exceeded on 100% of steps. The reason is a scale mismatch in the released configuration. The step score normalises a *mean* over layers and heads with min-max ranges recorded for *sums* at the response level, and the final range is very narrow ($[-0.100, -0.029]$). Typical values (PKS $\approx$2.5 against a range of 0–216; ECS $\approx$0.2) give a combined score near -0.007, far above the cut-off. In our faithful port, AARF therefore behaves as an always-on intervention rather than a targeted one.

C. Results

AARF changed the answer for 63 of 99 prompts; the other 36 were identical. Judged in the official order, AARF wins 33 pairs and the baseline 24 (6 verdicts unparsable), comparable to the modest gains the paper reports (Table XII). However, the judge reversed its verdict when the two answers were swapped in 22 of the 56 pairs where both verdicts were readable (39%). Counting only pairs where both orders agree, AARF wins 19 and the baseline 15, a difference well within chance (two-sided sign test $p = 0.61$). Much of AARF's apparent advantage in the official evaluation is therefore attributable to position bias. Its effect on truthfulness, in our setting, is not distinguishable from zero.

Other backbones. With each model's released hyperparameters, run on a Kaggle T4 GPU: Llama-2-13B: the gate fired on 100% of steps and changed 54 of 98 answers; with both orders required to agree, AARF won 14 and the baseline 12; Llama-3-8B: the gate fired on 8% of steps and changed 2 of 99 answers; with both orders required to agree, AARF won 0 and the baseline 0. The gate is thus either always on or almost never on, depending on the model's stored ranges, and in neither case does AARF measurably improve truthfulness.

XI. Discussion

What reproduces. The headline detection result reproduces closely on a laptop GPU, despite 4-bit weights and a sampled vocabulary for copying-head analysis. So do both mechanistic observations. The method also transfers to smaller and newer LLaMA models with little loss, which is encouraging for low-cost use.

What the pooled number hides. Per-task AUCs are 0.08–0.35 lower than pooled AUCs, and on data-to-text ReDeEP is unreliable for three of the four backbones. Evaluating hallucination detectors within task, or reporting both numbers, would give a more realistic picture of deployment performance.

Internals versus judges. Reading internals is cheap, local and needs no prompt, but on our data it is less accurate than asking a strong model. The two approaches fail differently. ReDeEP misses contradicted facts, while the judges are coarse and over-flag. A combination, for example ReDeEP as a free first-pass filter and a judge on flagged responses, is a natural next step.

XII. Limitations and Threats to Validity

- *Quantisation:* the 3B, 7B, 8B and 13B models ran in 4-bit NF4 (on the laptop or a free Kaggle T4), so their ECS and PKS approximate full precision. The close match with the paper and with the authors' logged scores suggests the effect is small.
- *Sampled copying-head scores:* OV circuits are scored on 500 sampled vocabulary tokens rather than the full vocabulary of 32k to 128k tokens.
- *One generator:* all responses come from llama-2-7b-chat, so the other three backbones score text they did not write. This differs from the paper's matched setting.
- *AARF judge:* AARF answers were judged by Gemini 3.5 Flash-Lite instead of GPT-4o, on Dolly (AC) only, as in the paper.
- *Dolly (AC) size:* 99 responses per backbone give fold standard deviations near 0.08–0.10 AUC.
- *Sample size:* 415 responses give fold standard deviations of 0.03–0.05 AUC, and differences below about 0.05 are within noise. Judge comparisons use 124 responses.
- *Robustness scope:* only numbers are corrupted. Entity swaps, negations and partially relevant passages are untested.
- *Judge access:* judges ran once, at default sampling settings, on free tiers. The GPT-family judge is the open-weights gpt-oss-120b, not a proprietary GPT-4-class model.

XIII. Conclusion and Future Work

We reproduced ReDeEP on a 6 GB laptop GPU. The detector reaches an AUC of 0.739 on Llama-2-7B against 0.7325 in the paper, both of the paper's mechanistic findings replicate, and on RAGTruth the method beats uncertainty baselines on all six LLaMA-family models we tested, from 1.1B to 13B parameters, with Llama-2-13B the strongest (AUC 0.801). Its accuracy is insensitive to the number of selected heads and layers, and both of its signals contribute.

With the authors' released code and data, chunk-level ReDeEP reproduces exactly (leakage-free AUC 0.731, identical to their own logged scores), and the method also works on Dolly (AC). The official evaluation selects features on the test set; this inflates results slightly on RAGTruth (0.016) and substantially, by about 0.08 on all three backbones, on the

small Dolly (AC) set. With the released settings, the AARF mitigation's gate is either always on (Llama-2) or almost never on (Llama-3-8B), and its benefit vanishes once the judge's position bias is controlled.

Our extension shows the method's boundary. ReDeEP catches responses that are off-topic for their context but not responses that contradict their context's numbers. Prompted judges from the Gemini and GPT families are more accurate, while small prompted judges do not work at all.

ReDeEP is therefore best used as a cheap, private first-line detector for topical hallucinations, paired with a fact-level check. Future work includes entity and negation corruptions, a numeric-consistency feature combined with ECS, full-precision runs on larger GPUs, a recalibrated AARF gate, and judge evaluation with repeated sampling.

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Appendix A
Hyperparameters

Setting	Value
Top-k attended context tokens	10% of context length
Copying heads $\lvert\mathcal{A}\rvert$ / FFN layers $\lvert\mathcal{F}\rvert$	8 / 5
Vocabulary sample for OV analysis	500 tokens, seed 0
Maximum sequence length	1,024 tokens
Maximum prompt + response length	6,000 characters
Cross-validation	stratified 5-fold, seed 0
Single split	70/30, seed 0 (124 test responses)
Classifier	logistic regression
Feature scaling	standardised (z-score)
Threshold	best F1 on training fold
Quantisation (3B, 7B)	bitsandbytes NF4, fp16 compute
Robustness corruption factor	uniform in $[1.3, 3.0]$

Appendix B
Judge Prompt

System message given to every judge:

```
You are an expert fact-checker for retrieval-
augmented generation. You will be given the input
that was shown to an AI assistant (a task
instruction plus retrieved reference content) and
the assistant's response. Decide whether the
response contains any hallucination: a claim that
contradicts the reference content, or information
that is not supported by the reference content.
Judge only against the reference content, not
against your own world knowledge.
```

The user message contains the full RAG input and the response, and asks for a JSON reply with two fields: `hallucination_probability`, an integer from 0 to 100, and a one-sentence `reason`.

Appendix C
Reproducibility

All code, cached scores and logs are in the project repository. From the `redeep/` directory:

```
pip install -r requirements.txt
bash scripts/run_llama_family.sh   # detection
bash scripts/run_extensions.sh     # robustness
bash scripts/run_api_judges.sh     # GPT/Gemini
python paper/analysis.py           # ablations etc.
python paper/build_paper.py        # this report
```

Every number in this report is generated from the result files by `build_paper.py`; none is typed by hand.